LAB-03 REPORT

Embedded Linux Debugging and Real-Time Performance Testing

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Course: Embedded Linux

Operating System: Ubuntu Linux

Platform: ARM QEMU / Cortex-A9

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1. INTRODUCTION

This laboratory focuses on Embedded Linux debugging, memory error detection, system call tracing, application performance analysis, and real-time scheduling evaluation.

2. OBJECTIVES

- Prepare an ARM debugging environment using QEMU.
- Analyze memory errors using Valgrind.
- Trace system calls using strace.
- Measure application performance using perf.
- Evaluate scheduling latency using cyclictest.
- Compare Standard Linux and PREEMPT_RT kernels.

3. EXPERIMENTAL ENVIRONMENT

Component	Description
Host OS	Ubuntu Linux
Architecture	x86_64
ARM Platform	Cortex-A9
Emulator	QEMU
Compiler	GCC / arm-linux-gnueabihf-gcc
Memory Analysis	Valgrind
Tracing	strace
Performance	perf
Real-time Test	cyclictest
RT Kernel	7.0.0-38-realtime

4. MEMORY ERROR DETECTION - VALGRIND

Valgrind was used to analyze memory errors in buggy_app_host. The application contains an out-of-bounds read in process_data(), a use-after-free in create_message(), and signed integer overflow in calculate().

Valgrind reported 47 errors from 8 contexts, including invalid and uninitialized memory accesses. The heap summary showed two allocations and two frees, with no outstanding heap blocks at exit.

5. SYSTEM CALL TRACING - STRACE

The strace utility was used to monitor system calls generated by the application. The trace included execve, openat, read, write, mmap, munmap, close and exit_group.

Result: logs/strace_log.txt

6. PERFORMANCE ANALYSIS - PERF

Metric	Result
Task Clock	0.54 ms
Context Switches	0

CPU Migrations	0
Page Faults	62
Elapsed Time	1.69 ms

The performance results are saved in logs/perf_stat.txt. This measurement was performed on the Ubuntu host.

7. REAL-TIME PERFORMANCE TESTING

Cyclictest was executed using FIFO scheduling, priority 80, a 1000 microsecond interval and 10000 iterations.

Metric	Standard	PREEMPT_RT
Minimum Latency	2 us	2 us
Average Latency	3 us	3 us
Maximum Latency	117 us	93 us
Iterations	10000	10000

The PREEMPT_RT kernel reduced maximum latency from 117 us to 93 us, an improvement of approximately 20.5 percent. Minimum and average latency remained unchanged.

The real-time kernel configuration was verified with CONFIG_PREEMPT_RT=y.

8. LIMITATIONS

Section A.2 (GDB Remote Debugging) was intentionally skipped. The perf measurement and cyclictest experiments were performed on the Ubuntu host. Results represent individual test runs and may vary under different system loads.

9. CONCLUSION

This laboratory demonstrated memory debugging with Valgrind, system call tracing with strace, performance measurement with perf, and real-time latency evaluation with cyclictest. The recorded results showed that PREEMPT_RT achieved a lower maximum scheduling latency than the Standard Linux kernel.